

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – TECHNICAL PRESENTATION

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 3 – Technical Presentation
Weightage:	10% of CIA	Total Marks:	100 (1 Question × 100 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	B.Shyam raj	Reg. No.:	192421176
Section / Batch:	2024	Date:	

Instructions to Students

- Answer the given question through a technical presentation. Total Marks: 100.
- Clearly define the problem and explain the concept of algorithm growth functions.
- Present comparative analysis using appropriate representations such as graphs, tables, or charts.
- Use suitable tools (PowerPoint, visualization tools, or software) to support your explanation.
- Ensure technical accuracy, logical flow, and proper interpretation of results.
- Maintain clarity in communication, proper slide organization, and professional delivery.
- Time management, confidence, and audience engagement will be considered during evaluation.

Question Type	Focus Area	Questions	Marks
Technical Presentation	Analyze and compare growth functions using mathematical techniques and asymptotic concepts and with graphical and visual interpretation	Q1	100
GRAND TOTAL:		100 Marks	

SECTION A – Technical Presentation

Q1. Divide and Conquer Recurrence Analysis with Logarithmic Combine Cost

Analyze a divide-and-conquer algorithm with recurrence $T(n) = 2T(n/2) + \log n$. Clearly define the subproblem structure and the additional combine work. Apply the Master Theorem to determine the time complexity and justify the selected case. Use diagrams or recursion trees to support the explanation. Interpret the result and justify the overall efficiency of the approach.

Code of Conduct

I B.Shyam raj(192421176) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Technical Content Accuracy	30	Highly accurate, in-depth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well-analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Marks Evaluation:

Question No.	Technical Content Accuracy (30%)	Problem Identification (25%)	Use of Tools (20%)	Communication (15%)	Professionalism (10%)	Total Marks (100)
Q1	4	4	4	2	3	90
Overall Total (100)						90

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

SLIDES:



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Divide and Conquer Recurrence Analysis with Logarithmic Combine Cost

Presented by:

B. Shyam raj

1924211176

CSA0620 : DESIGN AND ANALYSIS OF ALGORITHM

Dr. A. Sairam

Technical Presentation

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Introduction

- Divide and Conquer is a fundamental algorithm design technique.
- The problem is divided into smaller subproblems of equal size.
- Each subproblem is solved recursively.
- The solutions are combined to obtain the final result.
- The efficiency of such algorithms is analyzed using **recurrence relations**.
- In this analysis, the recurrence is:

$$T(n) = 2T(n/2) + \log n$$

where the combine step requires only logarithmic time.



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Core Technical Explanation

Divide and Conquer

Divide

Split the problem into **2 equal subproblems**.

Size of each subproblem = $n/2$

Conquer

Solve each subproblem recursively.

Combine

Merge or process the results.

Combine cost = $\log n$

General Form

$$T(n) = aT(n/b) + f(n)$$

Where

a = 2 (number of subproblems)

b = 2 (division factor)

f(n) = log n (combine cost)



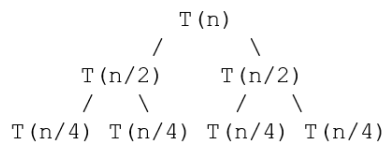
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Recursion Tree & Master Theorem

Recursion Tree



- The problem is divided into **2 equal subproblems** at each level.
- Each recursive call works on an input of size $n/2$.
- The recursion continues until the **base case ($n = 1$)** is reached.
- The height of the recursion tree is $\log_2 n$.



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REAL-WORLD APPLICATION

- Binary Search and recursive search algorithms
- Tournament scheduling
- Network routing
- Parallel computing
- Hierarchical file systems
- Cloud data processing



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Limitations & Future Scope

Limitations

- Recursive calls increase stack memory usage.
- Balanced partitioning is required for optimal performance.
- Recursive overhead affects very small inputs.
- Not suitable for problems that cannot be divided efficiently.

Future Scope

- Parallel Divide and Conquer algorithms
- GPU-accelerated recursive processing
- Hybrid algorithms combining Divide and Conquer with Dynamic Programming
- AI-based optimization of recursive algorithms
- Distributed computing applications



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Key Takeaways

Key Takeaways

- Divide and Conquer divides a problem into smaller subproblems and solves them recursively.
- The recurrence analyzed is: $T(n) = 2T(n/2) + \log n$.
- The logarithmic combine cost is much smaller than the recursive work.
- **Final Time Complexity:**

$$\Theta(n)$$



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Conclusion

- $T(n) = 2T(n/2) + \log n$ is an efficient **Divide and Conquer** recurrence.
- The **combine cost** ($O(\log n)$) is much smaller than the recursive computation.
- Using the **Master Theorem (Case 1)**, the overall time complexity is $\Theta(n)$.
- The algorithm scales efficiently and is suitable for solving **large computational problems**.
- The problem is recursively divided into **two equal subproblems** until the base case is reached.
- Its **linear time complexity** makes it efficient and scalable for processing large datasets.

GPS PHOTO:

